

Project Report Template

Advanced Analytics and Applications

Team 02



Authors:

Anthony Ge (7381058)

Hendrik Mehl (7410290)

Niklas Eichholz (7354459)

Yannick Herrmann (7409728)

Contact: student@smail.uni-koeln.de

Supervisor: Univ.-Prof. Dr. Wolfgang Ketter

Co-Supervisor: Janik Muires

Chair of Information Systems for Sustainable Society
Faculty of Management, Economics and Social Sciences
University of Cologne

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1 Problem description

The smart mobility market is highly competitive and current reports indicate that particularly ride-hailing, using autonomous and electric vehicles will dominate the market in the near future. Traditional car manufacturers face the risk of disruption if they fail to establish competitive mobility services. Our client, a renowned German car manufacturer, seeks to enter the ride-hailing market in the US, using a fully electrified fleet. However, the client lacks operational expertise in managing electric vehicle fleets in an urban ride-hailing context. Key challenges include understanding spatial and temporal demand patterns to deploy vehicles efficiently and optimizing charging schedules to reduce energy costs while ensuring vehicles are always sufficiently charged for their working day.

To address these challenges, we analyze historical taxi trip data from Chicago as a proxy for ride-hailing demand. Specifically, we aim to conduct a descriptive analysis, which analyzes taxi demand patterns and how they vary in different spatio-temporal resolutions. Second, we build predictive models that forecast taxi trip demand for a given spatial unit and time window, enabling the client to proactively position vehicles where and when they are needed. For example, this includes estimating expected demand on a rainy Sunday morning in the city outskirts. Furthermore, we develop an intelligent charging strategy using reinforcement learning to minimize charging costs while ensuring sufficient vehicle range. Together, these analyses will enable the client to make data-driven decisions regarding fleet deployment, driver scheduling, and charging infrastructure investment.

2 Data description

To approach this task, data from multiple sources was used. The primary dataset used in this analysis is the Chicago Taxi Trips dataset (City of Chicago, 2024), which is publicly available via the Chicago Data Portal and contains taxi trips from 2024 onwards. The Taxi Trip Dataset contains one trip per row, with timestamps rounded to the nearest 15 minutes. Since the full dataset starting from 2024 contains over 16 million rows, the dataset was limited to only include trips from 2025, resulting in a raw dataset of roughly 6.8 million rows. Important attributes include trip and taxi identifiers, the timestamp of the trip, centroid coordinates of the community area or census tract where the trip took place and multiple attributes about the trips properties like price or length (Figure 1).

Furthermore, hourly Chicago weather data was obtained from the Open-Meteo API (Open-Meteo, 2026) for the period of January 2024 to January 2026. The initial dataset includes roughly 17500 entries and key variables of the dataset include temperature, precipitation and sunshine duration (Figure 2).

Points-of-Interest data for Chicago was gathered from OpenStreetMap via the Overpass API (OpenStreetMap Contributors, 2026), covering nine categories: airports, train stations, stadiums, restaurants, bars and clubs, hotels, hospitals, universities, and tourist attractions. These raw POI locations are subsequently aggregated into distance and density features for multiple spatial resolutions during data preparation.

In order to be able to analyse the data on census tract and community area level, additional geometry data containing the shapes of these areas was obtained from . These datasets contain the geometry for Chicago's 77 community areas and 878 census tracts.

3 Data Preparation

Before analysis, each dataset underwent a series of preparation steps to ensure data quality and compatibility across sources.

3.1 Trip Data

The dataset was first inspected to get an overview of the datatypes and find potential issues. This revealed that several numeric columns were incorrectly stored as strings and that the timestamp columns were not yet in datetime format. Both were corrected during preprocessing. Additionally, the dataset contained no duplicated rows or trip identifiers. Furthermore, several implausible trips were identified and removed during cleaning. These included trips with zero or negative distance (528447 trips) and trips with a duration of less than 60 seconds (381488 trips), as these trips are likely erroneous. Overlapping trips with the same taxi identifier were also removed (28320 trips) based on the assumption that these result from recording errors. Price attributes were also validated by checking for and removing trips with zero or negative fares (1455 trips) and whether there are negative tips, tolls or extras (0 trips).

The dataset was also checked for missing values, revealing that 0.18% of the non-spatial data was missing. Due to the negligible share, these rows were dropped. Because of privacy reasons, the spatial data showed many more missing values with census tract data missing for approximately 55% of trips. In these cases, the less granular community area is recorded instead, meaning spatial information is always available at either the census tract or community area level [TRUE ???].

The numeric columns were checked for outliers using boxplots (Figure 3) and removed using the z-score as a statistical solution. Observations with a z-score exceeding a threshold of 2 in any numeric column were considered outliers and removed. [REASON]. This resulted in 444102 rows, or 7.18% of the data at that point being removed, which was seen as acceptable since the remaining set still contained roughly 5.7 million rows.

[CONTINUE]

3.2 Supplementary Data

4 Data Analytics

test4

4.1 descriptive analysis

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4.2 Prediction using Support Vector Machines

We modeled Chicago taxi demand using Support Vector Machines (SVM) at an H3 hexagon (resolution 6) 1-hour spatio-temporal resolution to optimize high-frequency fleet dispatch logistics.

To begin with, we had to select suitable features and preprocess the data. To prevent critical data leakage and establish strict structural boundaries, we excluded nine post-hoc operational and transaction variables that are only known after a trip concludes: `active_taxis`, `avg_idle_time`, `avg_trip_duration`, `avg_trip_distance`, `avg_fare`, `avg_trip_total`, `avg_tip`, `tip_rate`,

and `share_cash_payment`. Next, to avoid temporal leakage, the dataset of **289,080 entries** was partitioned using a strict chronological split (80% train / 20% test), reserving the final **57,816 intervals** for out-of-sample validation. Finally, the remaining predictors were routed through a `ColumnTransformer`, which normalized continuous features via `StandardScaler` to baseline variances and mapped categorical spatial tags (`pickup_h3_res6`) via `OneHotEncoder` to compute localized neighborhood intercepts.

To establish an initial understanding of feature importances, we conducted a linear coefficients analysis. This revealed that Point of Interest (POI) data strongly drives trip demand, led by density metrics such as `hotels_per_km2` (+18.55) and `attractions_per_km2` (+14.89). Conversely, local weather metrics like `apparent_temperature` (-0.89) and `rain` (+0.15) represent low-impact statistical noise.

As an initial benchmark, we evaluated a `LinearSVR` model without a kernel. This baseline underperformed, yielding an R^2 of **0.6134** (MAE = 13.05 trips). Its primary limitation was a severe boundary infraction: **34.52% of predictions fell below zero**, exposing the linear plane’s inability to respect non-negative demand thresholds.

Subsequent grid-search optimization (detailed in Figure 4) clearly identified the Radial Basis Function (RBF) kernel as the most promising approach. It also revealed 1000 as most promising value for the hyperparameter `C`. However, exact RBF kernel training scales cubically ($O(N^3)$) with dataset size, rendering standard training computationally intractable on our hardware. To bypass this bottleneck, we deployed a **Nystroëm low-rank approximation** to project the infinite-dimensional RBF space onto a compact set of landmark points. We evaluated different landmark counts alongside models with and without weather features, given that our previous linear coefficient analysis flagged weather metrics as marginal. As shown in Figure 5, our top-performing configuration excluded weather data entirely, climbing to **0.8734 R^2** and driving the MAE down to **6.49 trips**.

While these performance metrics represent a robust fit—especially given a highly efficient training runtime of approximately 7.5 minutes—the model still produced negative predictions for 22.28% of intervals, which required artificial post-hoc clipping to zero. To naturally enforce physical boundary constraints, we developed a decoupled **Two-Stage Hurdle SVM Framework**. Stage 1 (The Gatekeeper) utilizes an RBF-approximated `LinearSVC` binary classifier to map the active-demand perimeter (1 if active, 0 if dead). Stage 2 (The Volume Engine) then runs an RBF-approximated `LinearSVR` trained *exclusively* on active instances, shielding the regression plane from zero-floor distortions (see Figure 6 for the exact results).

Despite a minor structural tendency to generate false zeros during low-demand hours, this optimized **No Weather Hurdle Model emerged as our champion architecture, achieving an elite 0.8751 R^2 and a new floor of 6.44 trips MAE**. Crucially, it maximizes variance capture while mathematically guaranteeing **0% negative boundary violations**, rendering it highly viable for production deployment.

We also explored how performance scales across different spatio-temporal resolutions. Decreasing temporal resolution (aggregating from 1-hour to 6-hour and 24-hour windows) smooths high-frequency demand surges, naturally boosting variance capture. The **6-Hour Hexagon (No Weather)** model secured our highest raw fit at **0.9042 R^2** (MAE = 32.66 trips). However, absolute errors scale with the larger temporal footprint, which limits its utility for real-time dispatching.

Shifting the spatial unit from uniform geometric hexagons to irregular, administrative boundaries (Census Tracts/Community Areas) degrades true operational accuracy. The 24-Hour community area model yields a deceptively high **0.8929 R^2** but suffers a massive absolute error distortion of **57.12 trips MAE** (compared to the 24-Hour Hexagon’s **26.98 MAE**). This surge

in MAE is a textbook demonstration of the **Modifiable Areal Unit Problem (MAUP)**. It is driven primarily by the **Scale Effect**: because administrative Community Areas are larger on average ($\sim 7.68 \text{ km}^2$) than Resolution 7 hexagons ($\sim 5.16 \text{ km}^2$), they aggregate a much higher absolute volume of hourly trips, naturally inflating scale-dependent absolute metrics like MAE. Furthermore, the **Zone Effect** introduces irregular boundaries and mismatched spatial centroids, warping the distance metrics calculated on the SVM hyperplane. A full comparison of models can be found in Figure 7 and Figure 8.

Finally, we would try out these potential improvements in Follow-Up projects:

1. **Classifier Threshold Calibration:** Optimize the Hurdle’s Stage 1 gatekeeper to explicitly resolve the R^2 -destructive false-zero bottleneck.
2. **Spatial-Lag Graph Embeddings:** Integrate a spatial adjacency weight matrix (W) or graph neural network embeddings to capture continuous neighborhood demand overflow.
3. **Dynamic Landmark Sampling:** Reconfigure the Nystroëm approximation using a k-means clustering anchor selection rather than random sampling to optimize tracking across hyper-local urban corridors.

4.3 Analysis using Neural Networks

test

4.4 Analysis using Reinforcement Learning

test

5 Conclusion

conclusion text

6 Appendix

Columns (23)			
Column Name	Description	API Field Name	Data Type
 Trip ID	A unique identifier for the trip.	trip_id	Text
 Taxi ID	A unique identifier for the taxi.	taxi_id	Text
 Trip Start Timestamp	When the trip started, rounded to the nearest 15 minutes.	trip_start_timestamp	Floating Timestamp
 Trip End Timestamp	When the trip ended, rounded to the nearest 15 minutes.	trip_end_timestamp	Floating Timestamp
 Trip Seconds	Time of the trip in seconds.	trip_seconds	Number
 Trip Miles	Distance of the trip in miles.	trip_miles	Number
 Pickup Census Tract	The Census Tract where the trip began. For privacy, this Census Tract is not shown for some trips. This column often will be blank for locations outside Chicago.	pickup_census_tract	Text
 Dropoff Census Tract	The Census Tract where the trip ended. For privacy, this Census Tract is not shown for some trips. This column often will be blank for locations outside Chicago.	dropoff_census_tract	Text
 Pickup Community Area	The Community Area where the trip began. This column will be blank for locations outside Chicago.	pickup_community_area	Number
 Dropoff Community Area	The Community Area where the trip ended. This column will be blank for locations outside Chicago.	dropoff_community_area	Number
 Fare	The fare for the trip.	fare	Number
 Tips	The tip for the trip. Cash tips generally will not be recorded.	tips	Number
 Tolls	The tolls for the trip.	tolls	Number
 Extras	Extra charges for the trip.	extras	Number
 Trip Total	Total cost of the trip, the total of the previous columns.	trip_total	Number
 Payment Type	Type of payment for the trip.	payment_type	Text
 Company	The taxi company.	company	Text
 Pickup Centroid Latitude	The latitude of the center of the pickup census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	pickup_centroid_latitude	Number
 Pickup Centroid Longitude	The longitude of the center of the pickup census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	pickup_centroid_longitude	Number
 Pickup Centroid Location	The location of the center of the pickup census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	pickup_centroid_location	Point
 Dropoff Centroid Latitude	The latitude of the center of the dropoff census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	dropoff_centroid_latitude	Number
 Dropoff Centroid Longitude	The longitude of the center of the dropoff census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	dropoff_centroid_longitude	Number
 Dropoff Centroid Location	The location of the center of the dropoff census tract or the community area if the census tract has been hidden for privacy. This column often will be blank for locations outside Chicago. Read less ^	dropoff_centroid_location	Point

Figure 1: Raw Chicago Taxi Trip Data Column Descriptions

```

RangeIndex: 17568 entries, 0 to 17567
Data columns (total 15 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   date                                  17568 non-null  datetime64[ms, UTC]
1   temperature_2m                        17568 non-null  float32
2   relative_humidity_2m                 17568 non-null  float32
3   apparent_temperature                 17568 non-null  float32
4   precipitation                         17568 non-null  float32
5   rain                                 17568 non-null  float32
6   snowfall                             17568 non-null  float32
7   snow_depth                           17568 non-null  float32
8   surface_pressure                     17568 non-null  float32
9   cloud_cover                          17568 non-null  float32
10  wind_speed_10m                       17568 non-null  float32
11  wind_speed_100m                      17568 non-null  float32
12  is_day                               17568 non-null  float32
13  sunshine_duration                    17568 non-null  float32
14  direct_radiation                     17568 non-null  float32
dtypes: datetime64[ms, UTC](1), float32(14)

```

Figure 2: Weather Data Columns

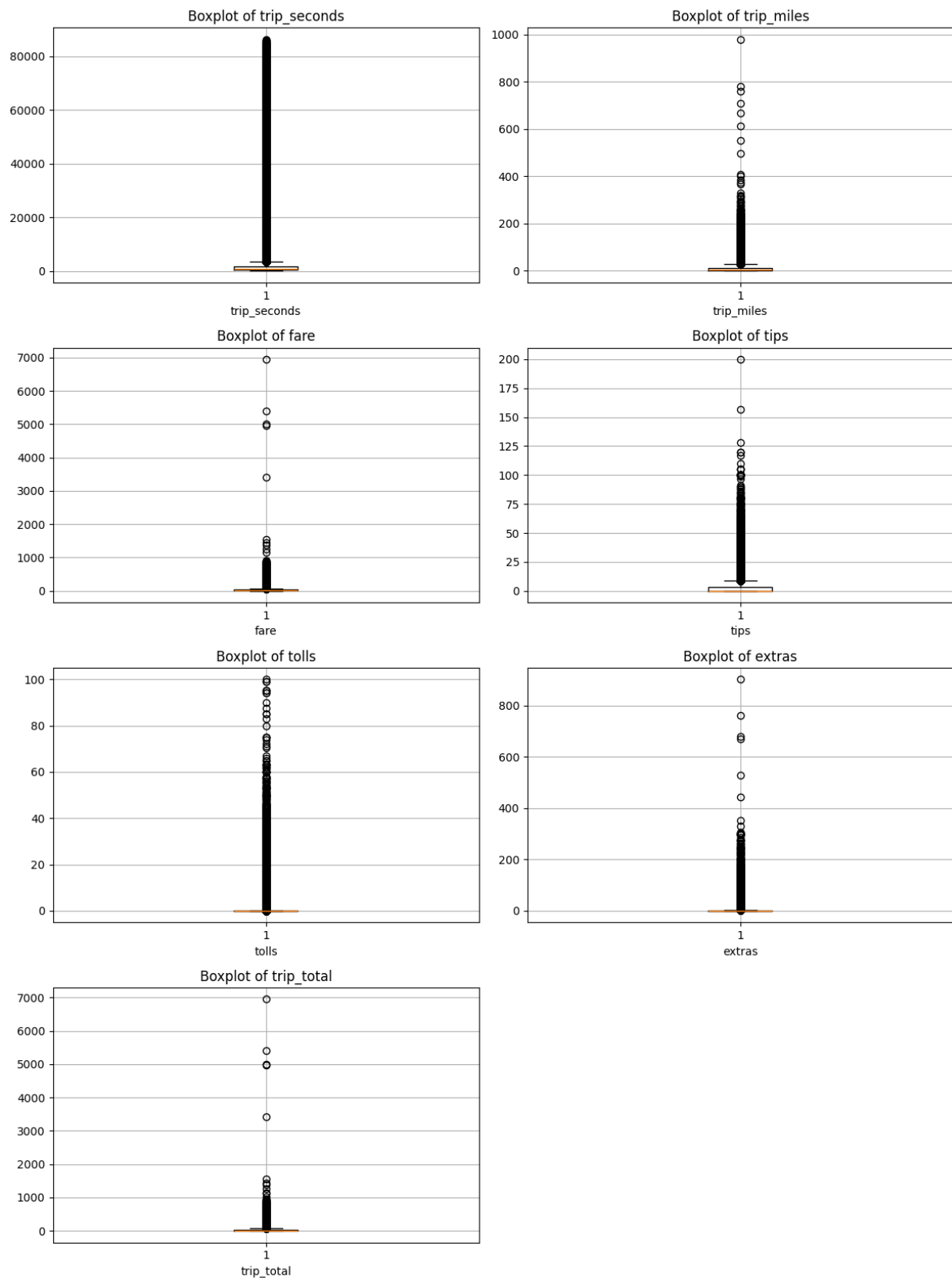


Figure 3: Boxplot of taxi trip attributes to detect outliers

Initiating full comparative Grid Search across 5000 rows...
 Fitting 3 folds for each of 18 candidates, totalling 54 fits
 Master Grid Search completed in 332.34 seconds!

=== THE GRADUAL COMPLEXITY KERNEL LEADERBOARD ===

param_kernel	param_C	param_degree	mean_test_score
rbf	1000	3	0.884359
rbf	1000	2	0.884359
poly	100	3	0.876292
rbf	100	2	0.873112
rbf	100	3	0.873112
poly	10	3	0.865926
poly	1000	2	0.857894
poly	100	2	0.842202
poly	1000	3	0.761338
poly	10	2	0.731756
rbf	10	2	0.586896
rbf	10	3	0.586896
linear	1000	3	0.490885
linear	1000	2	0.490885
linear	100	3	0.490872
linear	100	2	0.490872
linear	10	3	0.490677
linear	10	2	0.490677

Figure 4: Grid Search Results

Structural Configuration	Training Time	Out-of-Sample R^2	MAE (Trips)	Negative Predictions
Linear SVR Baseline (No Kernel)	0.42 seconds	0.6134	13.05	34.52% Outliers
Nystroëm RBF (1.5k, With Weather)	254.67 seconds	0.8316	8.30	28.55% Outliers
Nystroëm RBF (3k, With Weather)	2,164.74 seconds	0.8431	7.51	28.60% Outliers
Nystroëm RBF (1.5k, No Weather)	455.50 seconds	0.8734	6.49	22.28% Outliers

Figure 5: Model Results

```

=====
TWO-STAGE HURDLE SVM PERFORMANCE REPORT
=====
R-Squared ( $R^2$ ):                0.8751
Mean Absolute Error (MAE):         6.44 trips
Root Mean Squared Error (RMSE):    20.31 trips
Normalized RMSE (NRMSE):           112.23%
Mean Actual Test Demand:           18.10 trips
-----
Actual Zero-Demand Rows in Test:    20,723 / 57,816
Hurdle Predicted Zero Rows:         26,400 / 57,816
Negative Predictions Clipped:       0 (Mathematically Eliminated)
=====

```

Figure 6: Hurdle Results

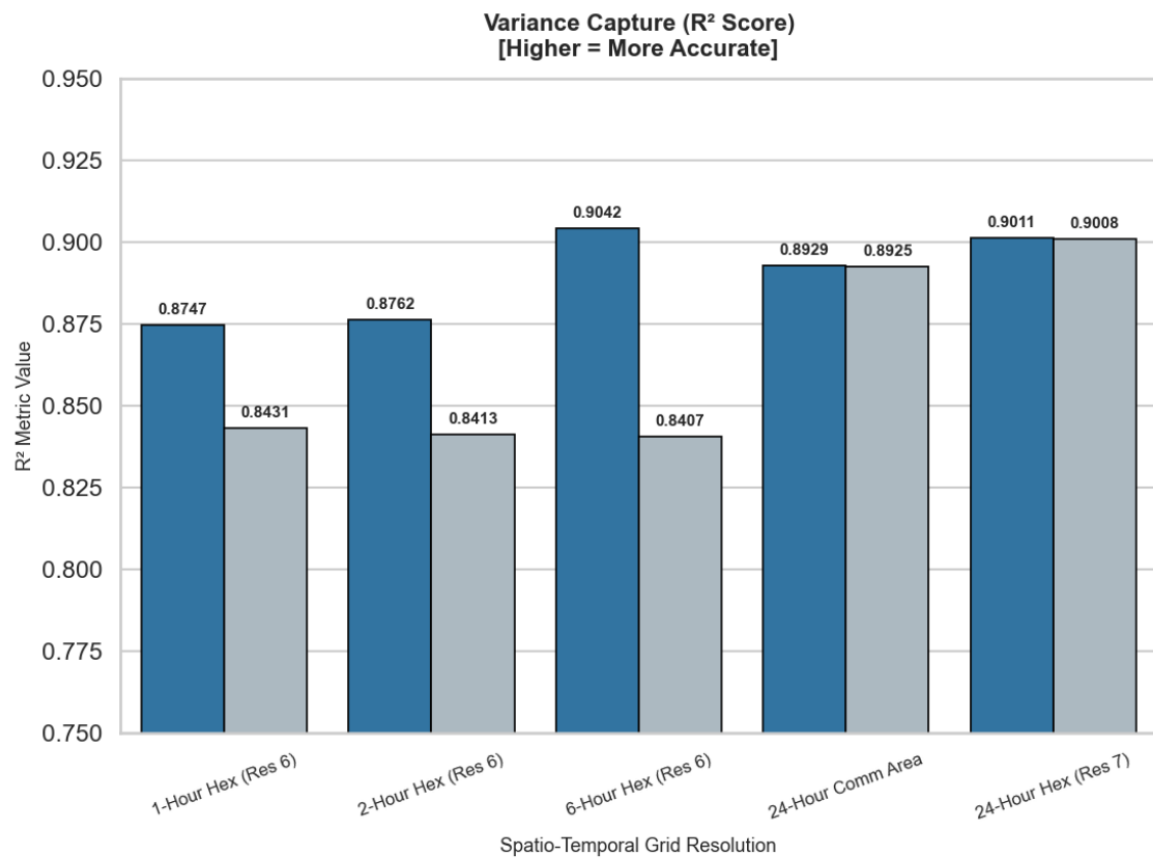


Figure 7: Spatio-temporal model comparison (R^2)

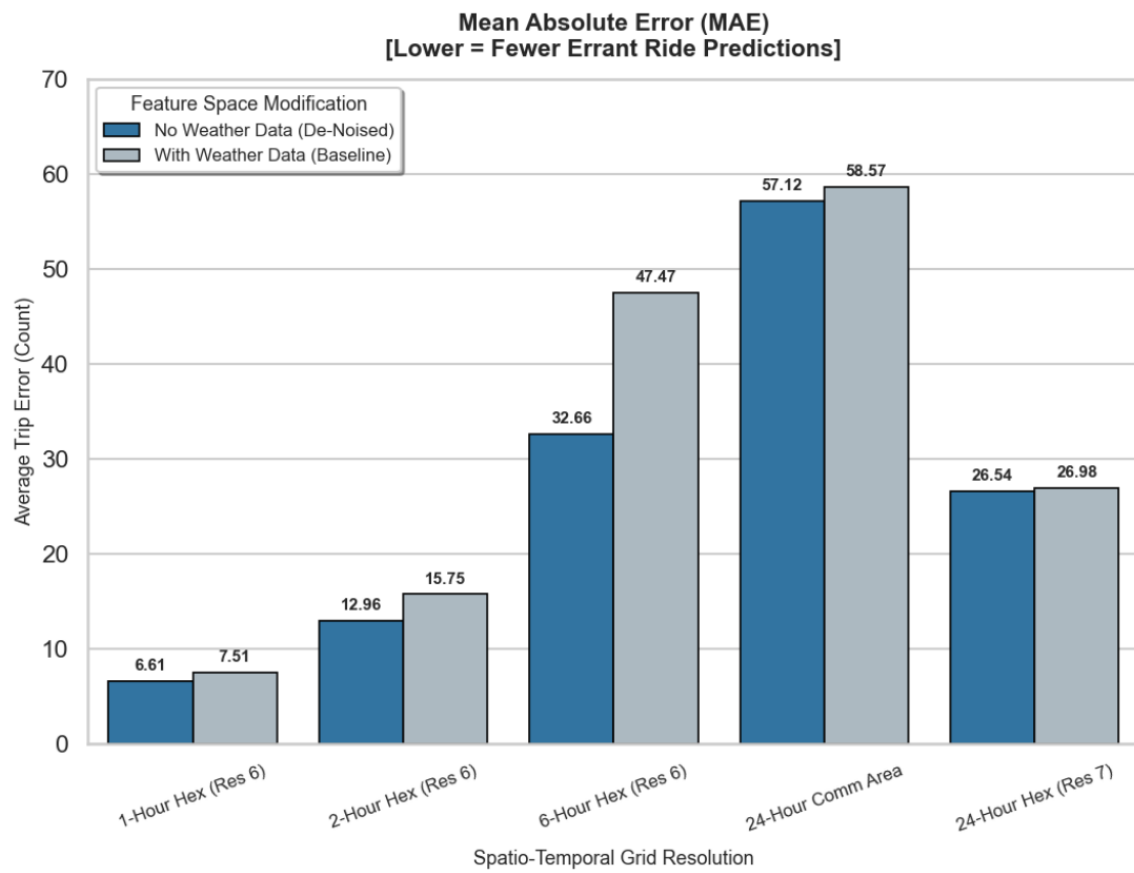


Figure 8: Spatio-temporal model comparison (MAE)

7 Citations and References

You can easily cite your sources using BibLaTeX. For example, you can cite the Quarto documentation (Team, 2024) or a textbook like “R for Data Science” (Wickham et al., 2023). The bibliography will be automatically generated at the end of the document.

References

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